

Final Project

Machine Learning for Big Data in Health Analytics
(BSTT 528)

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1 Introduction

Heart disease is the leading cause of death for men, women, and most ethnic groups in the United States. According to the CDC, one person dies every 34 seconds in the United States from Cardiovascular Disease. Heart disease costs the US around 216 billion yearly for health services and medicine. A large portion of heart disease is attributed to myocardial infarctions. A myocardial infarction, also known as a heart attack, occurs when there is a blockage that prevents blood from getting to the heart. For heart attacks specifically, someone has a heart attack every 40 seconds in the United States. Yearly, around 839,000 people in the United States have a heart attack and it accounts for around 1 in 3 deaths (Heart Disease and Stroke — CDC, n.d.). Around 1 in 5 heart attacks are silent, which means the person isn't aware of the heart attack. Risk factors for heart attacks include age, tobacco use, high blood pressure, high cholesterol, high triglycerides, obesity, diabetes, metabolic syndrome, family history, lack of exercise, unhealthy diet, stress, drug use, history of pre-eclampsia, or an autoimmune disease (Heart Attack - Symptoms and Causes - Mayo Clinic, 2022).

The government's response to this being the leading cause of death in the United States, the CDC created a Division for Heart Disease and Stroke Prevention which partners alongside government, public health, healthcare, and private sectors to improve prevention, detection, and control of heart disease focused specifically on cholesterol and high blood pressure. The Division for Heart Disease and Stroke Prevention funds heart disease and stroke prevention and management in all 50 states including some Native American Tribes in the US. Other efforts include WISEWOMAN which funds 21 states and 3 tribal organizations specifically focused on women aged 40 to 64 in low-income communities that have little or no access to health insurance. WISEWOMAN stands for Well Integrated Screening and Evaluation for Women Across the Nation. Heart disease is the leading cause of death in the United States accounting for 1 in 4 deaths of women (WISEWOMAN Home — cdc.gov, n.d.). There is also

Million Hearts which is a national initiative to prevent 1 million heart attacks and strokes in 5 years. The main focus of this organization is to make healthy communities, optimize care and focus on health equity (Million Hearts®, 2023).

Myocardial infarctions (MI) or heart attacks are the leading cause of death worldwide. Identifying factors that increase the risk of complications following a heart attack is essential to improving patient outcomes. There are various Machine Learning models that can identify patterns in predicting complications associated with myocardial infarctions. The analysis includes explorations of various Machine Learning techniques including Lasso, Elastic Net, Ridge, Random Forest, Boosting, and Classification Tree to predict the occurrence of complications resulting in death following myocardial infarction.

The application of Machine Learning has transformed health and clinical research. There is research on the Prediction of Incident Myocardial Infarction using Machine Learning Applied to Harmonized Electronic Health Data and the Prediction of Myocardial Infarction From Patient Features With Machine Learning. The prediction using Electronic Health Records (EHR) used data from 2.27 million patients in the UC Health system and used neural networks, grid search, and then compared a deep neural network to a simple logistic regression (Mandair et al., 2020). For the second one with predicting patient features, used various methods including XGBoost, gradient boosting, random forest, and support vector regression (Chen et al., 2022).

2 Data

The dataset comes from the UCI Machine Learning Repository, which became available in December 2020. This myocardial infarction dataset has 1700 patients with 124 attributes - 111 being features and 12 types of outcome variables which are complications including death. This dataset includes data from 1992 to 1995 in Krasnoyarsk, Russia.

The Myocardial Infarction Complications dataset aims to explore the relationship between various clinical and demographic factors associated with a myocardial infarction or a heart attack. The motivation behind choosing this dataset is to predict complications of myocardial infarctions based on the information of the patient. The dataset contains 1700 instances and 111 predictors including age, sex, Coronary heart disease, hypertension, and more. For this dataset, approximately 37% of these patients are female and 63% are male. The ages range from 26 to 92 with the average age being 61.9 with 8 patients without an age.

The outcome variable is binary and indicates whether the patient is still alive or died from a complication. Besides sex and age for demographic the rest of the predictor variables are related to health issues both past and at time at the hospital. Some of the predictors are dependent on time and length of hospital stay which is the time of admission, end of the first day, end of the second day, and end of the third day. If predicting after the end of the third day, we can consider all predictors which are how I approached this problem so we wouldn't lose any

of the predictors. Most of the predictor variables are either binary or ordinal. An example of an ordinal predictor is the presence of hypertension in which there is none and then three different stages of hypertension. The presence of heart failure is partially ordered with right, left, and both ventricular dysfunction. An example of a binary predictor would be the use of NSAIDS which is either yes or no. There are both complications following the myocardial infarctions which are variables 113 - 124 that include Atrial fibrillation, Pulmonary edema, Supraventricular tachycardia, Ventricular tachycardia, Ventricular fibrillation, Third-degree AV block, Myocardial rupture, Dressler syndrome, Chronic heart failure, Relapse of the myocardial infarction, Post-infarction angina and lethal outcome. Lastly, there is the lethal outcome which is binary but if it is lethal there are 7 possible lethal outcome categories. The complications are then broken into categories cardiogenic shock, pulmonary edema, myocardial rupture, heart failure, thromboembolism, asystole, or ventricular fibrillation.

The predictive variables besides age and sex include medical predictors ranging from chest pain (angina pectoris), ICU - intensive care unit, CHD - coronary heart disease, heart failure, electrocardiogram (ECG) , various types of blockages, qrs complex which is seen on an ECG or EKG, ESR - erythrocyte sedimentation rate in blood and the use of NSAIDs non-steroidal anti-inflammatory drugs. There are other types of illnesses listed in the predictive variables such as pneumonia, bronchitis, asthma, diabetes, obesity, and tuberculosis.

3 Data Preprocessing and Statistical Methods

The myocardial infarction dataset included a large number of missing values. Every patient had at least one missing value associated with a predictor variable. One predictor is chronic heart failure, which had different values 2 and 3 depending on if it was heart failure due to the left or right ventricular dysfunction. To keep these from being weighted differently in the dataset, these values between 2-3 were set to 2.5 to be of equal value since these should be counted as the same. To account for the missing values the mean imputation in R was used to compute the mean per column and replace those missing values with the mean. This is the preferred way because if we did the replacement with all zero values, that would skew the results since most of the predictors are binary so the only options are 0 or 1.

One of the most important data preprocessing steps in this analysis is the 10-fold cross-validation. Cross-validation is implemented by dividing the dataset into training and testing subsets. Then the model is trained on the training set and the model is evaluated on the testing set. With k-fold cross-validation, the dataset is split into 10 folds, where each fold is used as a testing set at some point. This will cycle through the 10 folds and use each fold as a test.

In supervised machine learning, an algorithm is trained on labeled data where data has been classified or categorized. Then the algorithm is trained to learn the patterns and relationships between the input features and the outcome variable. In supervised learning, there are two main sources for error in

the model which are bias and variance which leads to a Bias-Variance trade-off. A high bias will make the algorithm miss the relationships between the features and the target also known as underfitting. While the high variance will cause the model the noise which is overfitting. There are shrinkage methods or regularization which will shrink the coefficients in the model towards zero to reduce the variance. Some of these techniques learned in this course were Ridge Regression and Lasso Regression.

The first three statistical methods used in this analysis are Ridge Regression, Lasso Regression, and Elastic Net. Ridge Regression achieves a bias-variance tradeoff by reducing the model complexity. There are computational advantages for Ridge while also performing better when there is a large number of predictors. Ridge doesn't have the property of variable selection. Ridge is effective at reducing the variance of the estimates and is effective when the data have multicollinearity or the features are highly correlated. Lasso stands for Least Absolute Shrinkage and Selection Operator. With Lasso, the idea is similar to achieving a bias-variance tradeoff by minimizing β . For variable selection, Lasso shrinks some of β to zero. Lasso works well when there are a lot of features and we want to identify the most important features. Elastic Net is a combination of Lasso and Ridge by combining the penalties of both. Elastic net is favored when there are a large number of predictors and we want to perform both feature selection and regularization.

The other machine learning techniques used in this analysis are Classification and Regression Tree (CART) and Random Forest. CART is commonly used for both classification and regression problems. The basic idea of a regression tree uses binary splitting rules to separate subjects into different groups and use the mean response of each group for prediction. There are terminal nodes or leaves of the tree which are at the very bottom of the visualization. CART has many advantages, some being able to handle non-linear relationships between the input variables and the output. One possible downside of CART is overfitting if the tree is too complex but one way to account for this is by pruning the tree. Random Forest (RF) is a learning algorithm that combines multiple decision trees to improve the accuracy of the model. A random forest can be thought of as an extension of decision trees where it creates a set of decision trees and each is trained on a training set of data. Then the predictions from the original trees are combined to make a prediction. Random Forest uses boosting a technique to improve the accuracy and reduce the variance.

4 Results

The first three techniques were Ridge Regression, Lasso Regression, and Elastic Net. With all of these methods, the process using 10-fold cross-validation, fitting the model, finding the smallest lambda, re-running the model with the best lambda, calculating the ROC (Receiver Operating Characteristic Curve), and evaluating the performance of the model using AUC (Area under the ROC Curve). The AUC is a commonly used metric in Machine Learning for evaluating

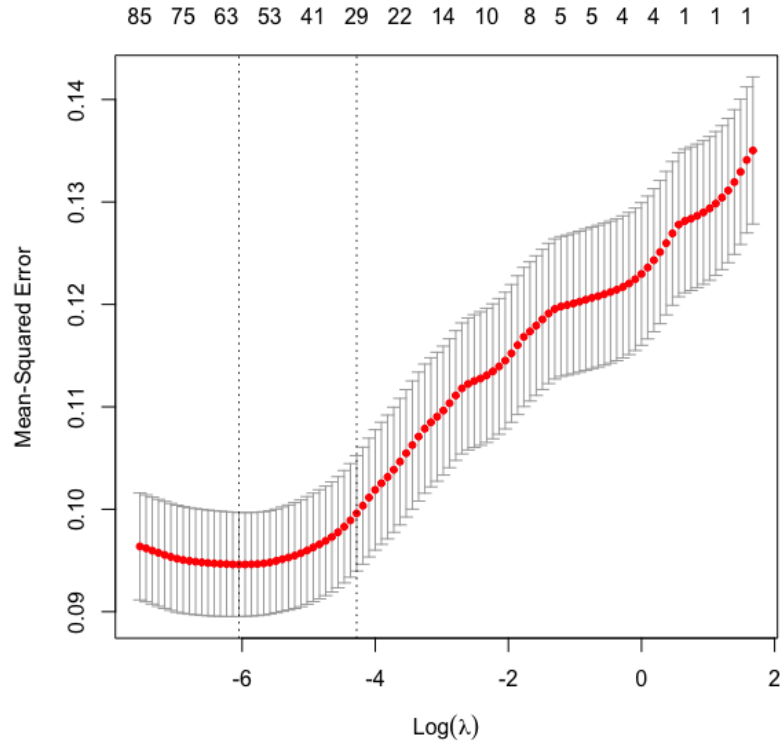


Figure 1: Elastic Net.

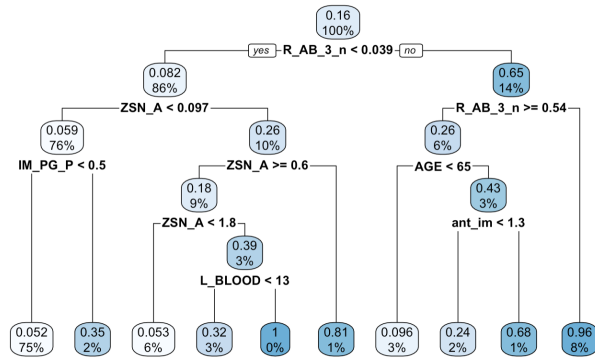


Figure 2: Pruned Classification Tree.

the performance of binary classification problems. The AUC measures the area under the ROC curve which gives a value to summarize the performance of the model.

For Ridge Regression, the AUC is 0.8088. For Lasso the AUC is 0.8372. For Elastic Net the AUC is 0.8376. The Elastic Net has performed the best repeatedly among these models. Elastic Net combines the strengths of both Lasso and Ridge by handling multicollinearity, a more generalizable model, better variable selection, and more flexibility. Elastic Net doesn't outperform Lasso by much though which should also be considered. Elastic Net did outperform Ridge, but that is expected as Ridge keeps all of the predictor variables and shrinks them but not all the way to zero, which can lead to overfitting if there are correlated predictor variables. A way to identify important predictor variables with a dataset with a large number of p is using SIS - Sure Independence Screening which is a feature selection method that identifies important features. SIS was implemented with a Lasso penalty which fits a Lasso Regression model with different parameters of the Lasso penalty parameter, but from using that method the Lasso model performed worse than the Lasso above using cross-validation with an AUC of 0.79.

The predictors that are common among all three of these models are. For Elastic Net specifically the top 8 predictors are cardiogenic shock in the ICU, right bundle branch block on the ECG when admitted to the hospital, right ventricular myocardial infarction, the use of liquid nitrates in the ICU, relapse of pain on the third day in the hospital, obesity present, bronchial asthma, and symptomatic hypertension. Similarly for Lasso, the top predictors are cardiogenic shock in the ICU, right ventricular myocardial infarction, right bundle branch block on ECG at the time admitted to the hospital, use of liquid nitrates in the ICU, relapse of pain the third day in the hospital, obesity, bronchial asthma, and obstructive chronic bronchitis, symptomatic hypertension and presence of chronic heart failure. These two models had a ton of overlapping predictors, but the ranking and the weight of each differed among Lasso and Elastic Net. The Ridge model predictors seem very different from the Lasso and Elastic Net models. The top predictors for Ridge are the Presence of an anterior myocardial infarction (left ventricular), the Presence of chronic Heart failure (HF), the Presence of an inferior myocardial infarction (left ventricular), the Use of opioid drugs in the ICU in the first hours of the hospital period, and Use of liquid nitrates in the ICU.

After these techniques, CART- Classification and Regression Tree and a Random Forest were implemented. Similar to the implementation of the models above, 10-fold Cross Validation was also used with CART. The top predictors from the CART model are relapsing in pain on the third day of the hospital, Presence of chronic Heart failure (HF) in the anamnesis, right ventricular myocardial infarction, age with a cutoff of younger than 65, Presence of an anterior myocardial infarction (left ventricular) and white blood cell count. For each of these predictors, similar to the age of younger than 65 there is a cutoff value per each of these predictors at varying levels of the classification tree. Similarly, AUC was used to measure the performance of this CART model, which was

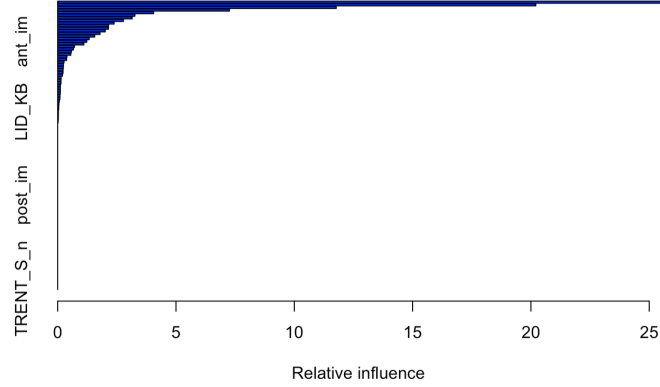


Figure 3: Boosting - Predictor Influence.

0.8156. This is in the same realm as the performance of the other models but the performance could be improved.

With the Random Forest, the analysis used 5-fold cross-validation and the number of trees set to 1000. Initially, the number of trees was 100, but the AUC was around 0.70, and decided to increase the number of trees to 1000 and the AUC increased only to about 0.73. Usually, the performance of the Random Forest would be higher than CART and the other models. The hyperparameters of the models were adjusted for the number of trees and the number of features for each split, but there wasn't much difference between tuning the parameters. When comparing the Random Forest to CART, the Random Forest is more sensitive to noise and can be prone to overfitting the data as well whereas CART is better at generalizing. Even without cross-validation and a 70/30 split on train and test data the Random Forest still performs about the same with an AUC right under 0.73.

In order to improve the performance of this model, Boosting was implemented from the GBM R package. Boosting is an approach to improve the predictions resulting from a decision tree. Bagging can be used in multiple instances in both classification and regression. For the input parameters, the number of trees stayed at 1000, and then for shrinkage used 0.01. With boosting, the AUC drastically improved to 0.887. This is the highest AUC from every model in this analysis. From the output of boosting, there is a metric in the summary relative influence which is a measure of each of the predictor variables in the model and indicates how much the variable contributes to the overall performance of the model. The predictors with the highest influence from boosting were the Use of NSAIDs in the ICU on the third day of the hospital period, the Use of opioid drugs in the ICU on the third day of the hospital period, Relapse of the pain on the third day of the hospital period, Presence of chronic

Heart failure (HF) in the anamnesis, Systolic blood pressure according to intensive care unit, Presence of a right ventricular myocardial infarction and Use of NSAIDs in the ICU in the second day of the hospital period. While some of these overlap with the predictors above, some of these are variables that weren't in the top 10-15 predictors of other models.

5 Conclusion

In conclusion, our analysis utilized several machine learning techniques including Lasso, Ridge, Elastic Net, Random Forest, Classification Tree, and Boosting to predict death from complications from myocardial infarctions. Elastic Net and Boosting outperformed other models in terms of performance and accuracy. The influence analysis from Boosting showed that use of NSAIDs, use of opioids, relapse of pain on the third day, chronic heart failure, systolic blood pressure for those in the ICU, right ventricular myocardial infarction, right bundle branch block on the ECG when admitted to the hospital, symptomatic hypertension and obesity were consistent among all models.

This analysis can impact clinical decision-making by making accurate predictions of death from myocardial infarctions which could potentially help physicians identify high-risk patients who require additional interventions. The findings show the importance of machine learning in analyzing medical data which provides valuable insights.

The next step for this analysis would be to carry out a similar analysis but for all of the outcome variables, there are 7 other outcome variables for complications not resulting in death. This analysis is limited though in sample size, since for the outcome variable used in this model all of the complications were combined that led to death which is 7 possible types. Ideally, there would be an analysis of each type of complication that led to death. Future studies should aim to validate our findings using datasets with a larger sample size. Overall, this study demonstrates the potential of machine learning models in predicting clinical outcomes and has important implications for improving patient care.

References

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2. Mandair, D., Tiwari, P., Simon, S. R., Colborn, K. L., Rosenberg, M. I. (2020). Prediction of incident myocardial infarction using machine learning

applied to harmonized electronic health record data. BMC Medical Informatics and Decision Making, 20(1). [BMC Medical Informatics and Decision Making](#)

3. Heart Disease and Stroke — CDC. (n.d.). [Heart Disease and Stroke — CDC](#).

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5. Million Hearts. (2023, April 14). Centers for Disease Control and Prevention. [Million Hearts](#).

6. UCI ML Dataset [UCI ML Dataset Myocardial Infarction](#)

6 Appendix

```
df_final <- data.frame(Myocardial)
df_final$LET_IS <- replace(df_final$LET_IS, df_final$LET_IS > 1, 1)
# 1429 no complications and 271 complications

df_final$ZSN_A <- replace(df_final$ZSN_A, df_final$ZSN_A == 2, 2.5)
df_final$ZSN_A <- replace(df_final$ZSN_A, df_final$ZSN_A == 3, 2.5)
##      0      1      2      3      4
## 1468  103   27   29   19

X = df_final[2:112]
Y = df_final$LET_IS

#Mean Imputation
df_fm <- df_final
for (i in 1:ncol(df_fm)) {
  df_fm[, i][is.na(df_fm[, i])] <- mean(df_fm[, i], na.rm = TRUE)
}

library(leaps)
library(glmnet)
library(data.table)
library(coefplot)
library(pROC)
library(ggplot2)

df_f <- df_fm
x = df_f[2:112]
#the last outcome being lethal outcome
```

```
y = df_f$LET_IS
```

```
dat1 = data.frame(y,x)
## nested CV
cv_err = rep(0, 3)
nfold = 10
foldid = sample(rep(1:nfold, length=nrow(dat1)))

for(k in 1:nfold){
  dat_train = dat1[foldid !=k,]
  dat_test = dat1[foldid ==k,]
  x=as.matrix(x)

  ## Ridge
  fit_ridge = cv.glmnet(x=x[foldid !=k,],
                        y=y[foldid !=k,],
                        alpha=0, standardize = FALSE)
  best_lambda_r <- fit_ridge$lambda.min
  best_model_r <- glmnet(x=x[foldid !=k,],
                        y=y[foldid !=k,], alpha =0, lambda = best_lambda_r, standardize = FALSE)
  coef_r = coef(best_model_r)[-1,1]
  y_hat_1 = predict(best_model_r, newx=x[foldid ==k,], s = best_lambda_r)
  y_hat_1[foldid ==k] = y_hat_1
  # check the AUC
  roc_ridge = roc(y, y_hat_1)
  ggroc(roc_ridge)
  auc(roc_ridge)

  ## Lasso
  # extract.coef(res_lasso)

  ## Lasso
  res_lasso = cv.glmnet(x=x[foldid !=k,],
                        y=y[foldid !=k,], alpha=1, standardize = FALSE)
  best_lambda <- res_lasso$lambda.min
  best_model <- glmnet(x=x[foldid !=k,],
                        y=y[foldid !=k,], alpha =1, lambda = best_lambda, standardize = FALSE)
  coef_lasso = coef(best_model)[-1,1]
  #coef_lasso = coef(res_lasso)[-1,1]
  y_hat_2 = predict(best_model, newx=x[foldid ==k,], s = best_lambda)
  y_hat_2[foldid ==k] = y_hat_2
  # check the AUC
  roc_lasso = roc(y, y_hat_2)
  ggroc(roc_lasso)
  auc(roc_lasso)
```

```

## Elastic Net, fix alpha=0.5
res_enet = cv.glmnet(x=x[ foldid !=k, ],
                    y=y[ foldid !=k ], alpha=0.5,
                    standardize = FALSE)
coef_enet = coef(res_enet)[-1,1]
best_lambda_e <- res_enet$lambda.min
best_model_e <- glmnet(x=x[ foldid !=k, ],
                    y=y[ foldid !=k ], alpha =0.5, lambda = best_lambda_e, stand
plot(res_enet)
coef_e = coef(best_model_e)[-1,1]
y_hat_3 = predict(best_model_e,newx=x[ foldid==k,])
y_hat_3[ foldid==k ] = y_hat_3
# check the AUC
roc_enet = roc(y, y_hat_3)
ggroc(roc_enet)
auc(roc_enet)
#sort(coef_e[coef_e[,1]!=0,])
}

```

```

# CART
library(rpart)
library(rpart.plot)
library(randomForest)
library(factoextra)

```

```

# full dataset
fit_cart = rpart(y ~ .,data=dat1, control = rpart.control(minsplit = 5))
#summary(fit_cart)
rpart.plot(fit_cart)

```

```

library(tree)
fit_cart2 = tree(y~.,data=dat1, control=tree.control(nobs=nrow(dat1), mincut = 5)
fit_cart_cv = cv.tree(fit_cart2,K=10)
fit_cart_cv$dev
fit_cart_cv$size

```

```

min.ind = which.min(fit_cart_cv$dev)
min.size = fit_cart_cv$size[min.ind]
fit_cart_prune = prune.tree(fit_cart2, best=min.size)
plot(fit_cart_prune)
text(fit_cart_prune,pretty = 0)

```

```

## out-of-sample predicted outcome
dat1 = data.frame(y,x)
## nested CV
cv_err = rep(0, 3)
nfold = 10
foldid = sample(rep(1:nfold , length=nrow(dat1)))
yhat = rep(NA, nrow(x))

for(k in 1:nfold){
  dat.train = dat1[foldid!=k,]
  dat.test = dat1[foldid==k,]
  fit_cart_k = tree(y~., data=dat.train , control=tree.control(nobs=nrow(dat.train))
  fit_cart_k_cv = cv.tree(fit_cart_k, K=10)
  min.ind = which.min(fit_cart_k_cv$dev)
  min.size = fit_cart_k_cv$size[min.ind]
  fit_cart_k_prune = prune.tree(fit_cart_k, best=min.size)
  y_hat_k = predict(fit_cart_k_prune, dat.test)
  #y_hat_k = predict(fit_cart_k_prune, dat.test)[,2]
  yhat[foldid==k] = y_hat_k
}
#roc(y, y_hat_k)
#AUC
roc_tree = roc(y, yhat)
ggroc(roc_tree)
auc(roc_tree)

# RANDOM FOREST
dat1 = data.frame(y,x)
## nested CV
nfold = 5
foldid = sample(rep(1:nfold , length=nrow(dat1)))
yhat = rep(NA, nrow(x))
dat_train$y <- as.factor(dat_train$y)
dat_test$y <- as.factor(dat_test$y)
dat1$y <- as.factor(dat1$y)

for(k in 1:nfold){
  dat.train = dat1[foldid!=k,]
  dat.test = dat1[foldid==k,]
  # tried 100 trees and auc was 70
  # increasing trees to 1000 improved auc to 0.7297
  fit_rf_k = randomForest(y~., data=dat.train , importance =TRUE, mtry=5, ntree=1000)
  y_hat_rf = predict(fit_rf_k, dat.test)
  y_hat_rf[foldid==k] = y_hat_rf
}

```

```

plot(fit_rf_k)
## AUC
roc_tree = roc(as.numeric(dat1$y), as.numeric(y_hat_rf))
ggroc(roc_tree)
auc(roc_tree)

head(importance(fit_rf_k))
varImpPlot(fit_rf_k)

#BOOSTING
library(gbm)

# out-of-sample predicted outcome
dat1 = data.frame(y,x)
## nested CV
nfold = 5
foldid = sample(rep(1:nfold, length=nrow(dat1)))
yhat = rep(NA, nrow(x))
dat_train$y <- as.factor(dat_train$y)
dat_test$y <- as.factor(dat_test$y)
dat1$y <- as.factor(dat1$y)

for(k in 1:nfold){
  dat.train = dat1[foldid!=k,]
  dat.test = dat1[foldid==k,]
  # tried 100 trees and auc was 70
  # increasing trees to 1000 improved auc to 0.7297
  fit_rf_b = gbm(y ~., data=dat.train,
                 distribution="gaussian",
                 n.trees=1000,
                 interaction.depth=1,
                 shrinkage = 0.01)
  y_hat_b = predict(fit_rf_b, dat.test)
  y_hat_b[foldid==k] = y_hat_b
}

plot(fit_rf_b)
## AUC
roc_tree_b = roc(as.numeric(dat1$y), as.numeric(y_hat_b))

ggroc(roc_tree_b)
auc(roc_tree_b)

head(summary(fit_rf_b, plotit=T))

```